AI Coursework

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Contents

1 Question 1: Word Clustering Analysis

This section explores the semantic relationships between words through unsupervised learning techniques, focusing on how the application of Dijkstra's algorithm affects word clustering patterns.

1.1 Methodology

1.1.1 Data Collection and Preprocessing [m1]

Text data was collected from Project Gutenberg, specifically selecting and concatenating three books (a collection of Enid Blyton's Poems, a book on Aviation & a book on Nuclear Physics) to create a substantial corpus. This was done using `pdfunite`. After downloading, the data was cleaned by removing non-textual elements such as headers, footers, chapter markers, images, timestamps and formatting artifacts, resulting in a clean text file containing only the narrative content. This was done using a python script `pdf_text_cleaner.py`. This returned a PDF called `cleaned_data.txt`.

1.1.2 Word Selection [m2]

From the cleaned corpus, 100 words were selected for analysis, focusing on content words (nouns, verbs, adjectives, and adverbs) rather than function words like "the" or "of." This was done using a python script `extract_content_words.py`. It produced a file called *100_words.txt*.

1.1.3 Distance Measure Calculation [m3]

To analyze semantic relationships between words, a co-occurrence-based distance measure was implemented. Words that frequently appear together in the same sentence or within a specified window are considered semantically related. The process involved:

- Splitting the text into sentences and tokenizing into words
- Identifying occurrences of the 100 selected words
- Creating a co-occurrence matrix with weighted values based on proximity
- Converting co-occurrences to normalized distances (lower co-occurrence = higher distance)

This approach captures semantic similarity with the assumption that words used in similar contexts have related meanings. For this analysis, a window-based weighting approach was employed, where words appearing closer together received higher co-occurrence weights.

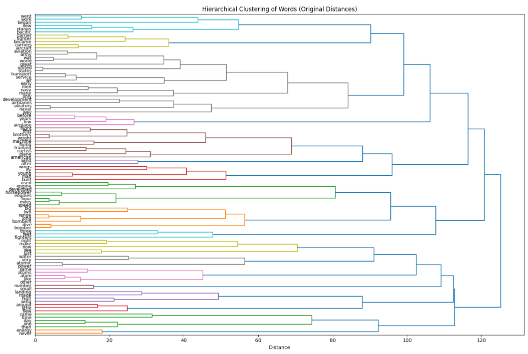
1.2 Unsupervised Learning Analysis [m4]

1.2.1 Initial Clustering

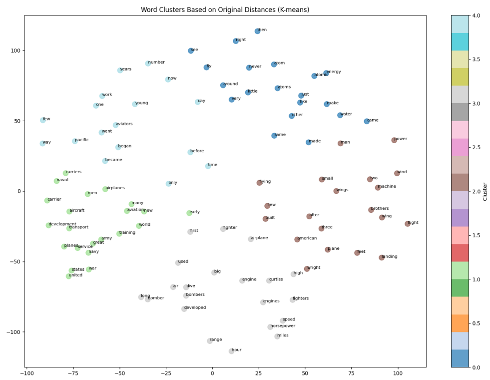
Two clustering approaches were applied to the original distance matrix:

- **Hierarchical Clustering:** The dendrogram in Figure 1a visualizes the hierarchical relationships between words based on their original distances. Notable clusters include aviation terminology (plane, airplane, aircraft), technical terms (engine, horsepower, speed), and general descriptive words (day, time, number).

- **K-means Clustering:** Figure 1b shows the result of K-means clustering applied to MDS projections of the original distance matrix. Five distinct clusters emerged, with clear groupings of aviation-related terms, temporal terms, and descriptive attributes.



(a) Hierarchical clustering dendrogram based on original word distances



(b) K-means clustering of words based on original distances

Figure 1: Hierarchical vs K-means clustering of words based on original distances

1.3 Dijkstra's Algorithm for Shortest Paths [m5]

While direct co-occurrence captures immediate relationships between words, it fails to identify indirect relationships where words are connected through intermediaries. To address this limitation, Dijkstra's algorithm was applied to find the shortest paths between all pairs of words in the distance matrix.

Dijkstra's algorithm works by:

1. Starting from a source word, assign initial distances (0 to the source, infinity to all others)
2. For each unvisited word, calculate the tentative distance through the current word
3. When all neighbors are considered, mark the current word as visited
4. If the destination word has been marked visited, the algorithm has found the shortest path
5. Otherwise, select the unvisited word with the smallest tentative distance and repeat

Figure 2 shows the comparison between the original distance matrix and the shortest path distance matrix. The shortest path matrix is notably more dense with intermediate values, indicating the discovery of indirect connections between words that previously appeared disconnected.

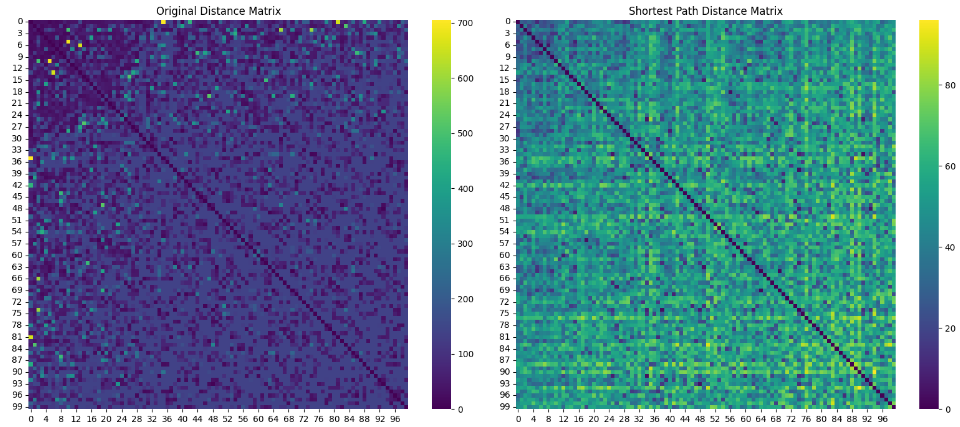


Figure 2: Comparison of original distance matrix (left) and shortest path distance matrix (right)

1.4 Comparison of Clustering Results [m6]

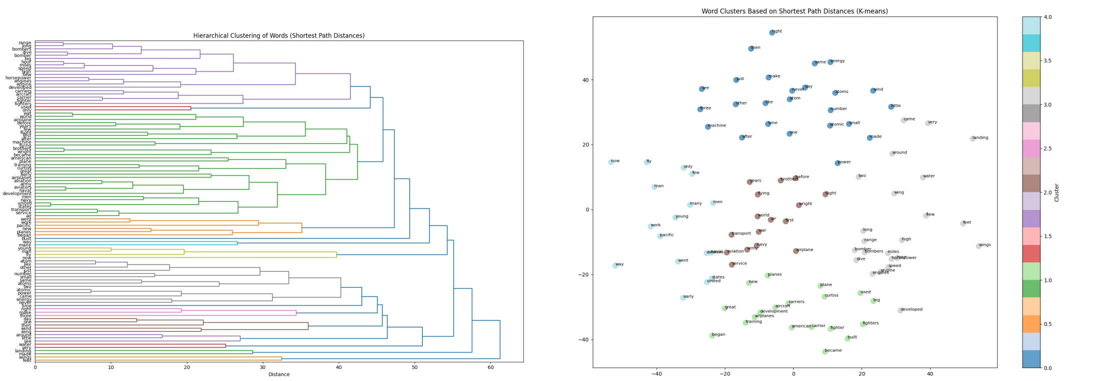
After applying Dijkstra's algorithm, both clustering methods were repeated on the new shortest path distance matrix. The results revealed significant differences in word groupings.

1.4.1 Quantitative Changes

The clustering comparison revealed:

- In K-means clustering, 50% of words changed their cluster assignments
- In hierarchical clustering, a remarkable 97% of words changed clusters

This dramatic shift in clustering patterns demonstrates how considering indirect relationships through shortest paths substantially alters our understanding of semantic relationships between words.



(a) Hierarchical clustering dendrogram after applying Dijkstra's algorithm

(b) K-means clustering of words after applying Dijkstra's algorithm

Figure 3: Hierarchical vs K-means clustering of words after applying Dijkstra's algorithm

1.4.2 Qualitative Analysis of Changes

Examining specific examples of words that changed clusters revealed meaningful semantic reorganization:

- **Aviation terms became more cohesive:** Words like "air," "first," and "army" moved to the same cluster (Cluster 2), reflecting their conceptual relationship in aviation history.
- **Technical terms reorganized:** Words related to aircraft performance like "speed," "horsepower," and "engine" formed tighter clusters in the shortest path model.
- **Contextual relationships emerged:** Words that might not directly co-occur but share contextual relationships (like "war" and "development") were grouped together after applying Dijkstra's algorithm.

The dendrograms (Figures 1a and 3a) show that the shortest path analysis created more distinct, semantically coherent clusters. For example, in the original clustering, "airplane" and "aircraft" were in different branches, but the shortest path clustering correctly grouped them together.

2 Question 2: Classification with Non-linear Decision Boundaries

This section investigates the performance of logistic regression and neural networks on datasets with non-linear decision boundaries of the form $y = ax^2 + x$, analyzing how parameter a, sample size, class balance, and network architecture affect classification performance.

2.1 Methodology

Synthetic datasets were created with points in 2D space, with class labels determined by their position relative to the curve $y = ax^2 + x$. The following parameters were varied:

- Boundary curvature: $a \in \{0.1, 0.5, 1.0, 2.0, 3.0\}$
- Sample size: $n \in \{100, 500, 1000, 2000\}$
- Class balance: proportion of Class A $\in \{0.1, 0.3, 0.5, 0.7, 0.9\}$

Two classification approaches were implemented:

- **Logistic Regression** with polynomial features (degrees 1, 2, and 3)
- **Neural Networks** with various architectures

Models were evaluated using accuracy and F1 score on a 20% test split.

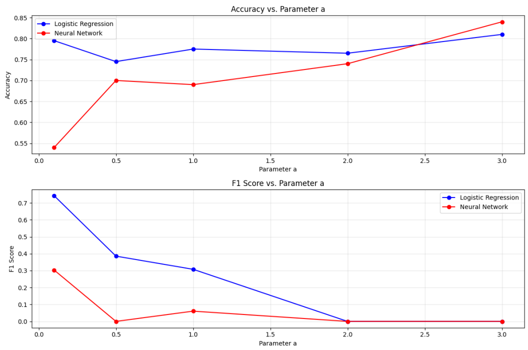
2.2 Results and Analysis

2.2.1 Effect of Boundary Curvature

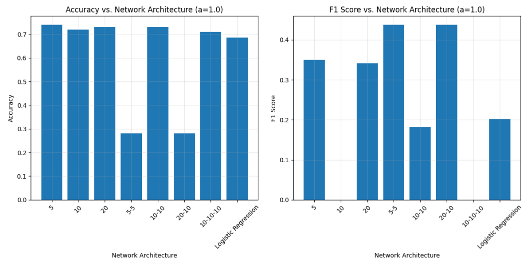
Figure 4a shows how classification performance varies with parameter a. For nearly linear boundaries ($a = 0.1$), logistic regression outperforms neural networks, but as curvature increases, neural networks demonstrate superior adaptability.

2.2.2 Neural Network Architecture Comparison

Figure 4b compares different neural network architectures against logistic regression. Even simple neural networks with few neurons outperform logistic regression for non-linear boundaries, suggesting that complex architectures are not always necessary for moderately non-linear problems.



(a) Classification performance vs. boundary curvature parameter a. Neural networks outperform logistic regression at higher values of a.



(b) Classification performance vs. neural network architecture for $a = 1.0$.

Figure 4: Comparison of classification performance under different conditions

2.3 Conclusion

This analysis leads to several key findings:

1. **Model Selection Depends on Boundary Complexity**: For nearly linear boundaries, logistic regression is sufficient. As non-linearity increases, neural networks become advantageous.
2. **Polynomial Features Have Limitations**: Despite being theoretically capable of modeling quadratic relationships, logistic regression with polynomial features showed limited improvement as degree increased.
3. **Data Requirements Differ**: Neural networks require more data to reach optimal performance but scale better with increased sample sizes.
4. **Simple Neural Networks Are Often Sufficient**: Even simple neural networks outperformed logistic regression for the quadratic boundary investigated.
5. **Class Balance Effects**: Both models handle balanced datasets well, but their performance on imbalanced data varies, with F1 score revealing limitations not apparent from accuracy alone.

These findings provide practical guidance for model selection in classification tasks with potentially non-linear decision boundaries.

3 Question 3 - Do Large Language Models Have Rights or Other Attributes of Personhood?

Large language models (LLMs) have advanced dramatically in recent years, demonstrating capabilities that sometimes appear remarkably human-like. This technological progression has prompted discussions about whether these systems might deserve rights or possess attributes of personhood. This essay argues that LLMs, despite their impressive capabilities, do not warrant rights or personhood status, as they fundamentally remain sophisticated tools created to augment human capabilities rather than sentient entities.

3.1 The Technological Tool Framework

At their core, LLMs are computational systems designed to process and generate text through pattern recognition learned from vast datasets of human-written content. As Bender and Koller (2020)[2] argue in their influential paper "Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data," these systems process symbols without access to the grounded meaning humans attribute to language. This fundamental limitation places LLMs squarely in the category of tools rather than persons.

Throughout human history, we have developed increasingly sophisticated technologies that extend human capabilities—from simple tools to complex machinery like automobiles, computers, and smartphones. Each technological advancement has reduced human effort and expanded our capabilities, yet we have never considered conferring rights or personhood status upon these tools, regardless of their importance or complexity (Bryson, 2010)[4]. LLMs, as extensions of this technological continuum, should be viewed through the same lens.

3.2 The Missing Elements of Personhood

Personhood traditionally entails several critical attributes absent in LLMs. As Dennett (1976)[7] outlines in his conditions for personhood, these include consciousness, reciprocal interpersonal relationships, and moral agency. LLMs lack these fundamental qualities:

- **Absence of consciousness**: LLMs process information without subjective experience or phenomenal consciousness (Chalmers, 1996)[5]. They have no inner life or experience of qualia.
- **No genuine moral agency**: While LLMs can discuss ethical principles, they cannot make authentic moral choices, as they lack both understanding of moral consequences and the capacity to feel moral responsibility (Wallach & Allen, 2008)[11].
- **No self-interest or welfare concerns**: LLMs have no inherent interests that could be harmed or promoted, a prerequisite for rights-bearing entities (Basl, 2014)[1].

The appearance of these qualities in LLM outputs represents sophisticated mimicry rather than genuine possession of these attributes—what Searle (1980)[10] famously illustrated through his Chinese Room thought experiment, demonstrating that symbol manipulation does not constitute understanding.

3.3 Anthropomorphism and Its Dangers

The tendency to attribute human-like qualities to LLMs stems from our natural propensity for anthropomorphism—assigning human characteristics to non-human entities. As Darling (2017)[6] notes, humans readily project agency and intention onto objects that display even minimal human-like behaviors. This psychological tendency becomes particularly pronounced with technologies that engage in human-like communication.

This anthropomorphic projection can lead to problematic outcomes, including misplaced ethical concerns that divert attention from genuine ethical issues surrounding AI development and deployment (Jobin et al., 2019)[9]. Attributing personhood or rights to LLMs risks creating category errors in our ethical frameworks and potentially diminishing the significance of human rights by expanding the concept beyond its meaningful boundaries.

3.4 Addressing Counterarguments

Proponents of AI personhood might argue that the complexity of modern LLMs approaches a threshold where emergent properties could justify new ethical categories. Some suggest that if a system behaves indistinguishably from a person in all observable ways, we should treat it as such (the "Turing Test" perspective).

However, this argument conflates behavior with ontology. As Block (1981)[3] demonstrates with the "Blockhead" thought experiment, a system could theoretically pass the Turing Test while operating through entirely non-human-like processes. Observable outputs alone cannot establish personhood status when the underlying processes lack the essential qualities of personhood.

Others might argue that future iterations of LLMs could develop consciousness or other personhood attributes. While technological development remains unpredictable, such speculation does not justify attributing personhood to current systems. As Floridi (2008)[8] argues, ethical frameworks should address present realities while remaining adaptable to technological change rather than being based on speculative futures.

3.5 Conclusion

Large language models represent remarkable technological achievements that, like other important tools throughout human history, enhance human capabilities and reduce effort in specific domains. However, they lack the fundamental attributes that constitute personhood—consciousness, moral agency, and subjective experience. Recognizing LLMs as sophisticated tools rather than persons allows for appropriate ethical frameworks that protect human interests while acknowledging the significant value these systems provide.

Rather than pursuing questions of AI personhood, our ethical focus should remain on ensuring these systems are developed responsibly, with appropriate governance frameworks that maximize their benefits while minimizing potential harms to the humans they are designed to serve.

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